

4 Figure 4.1 shows an oxygen dissociation curve of adult haemoglobin.

The curve is obtained by measuring the percentage saturation of haemoglobin with oxygen at different partial pressures of oxygen under standardised conditions.

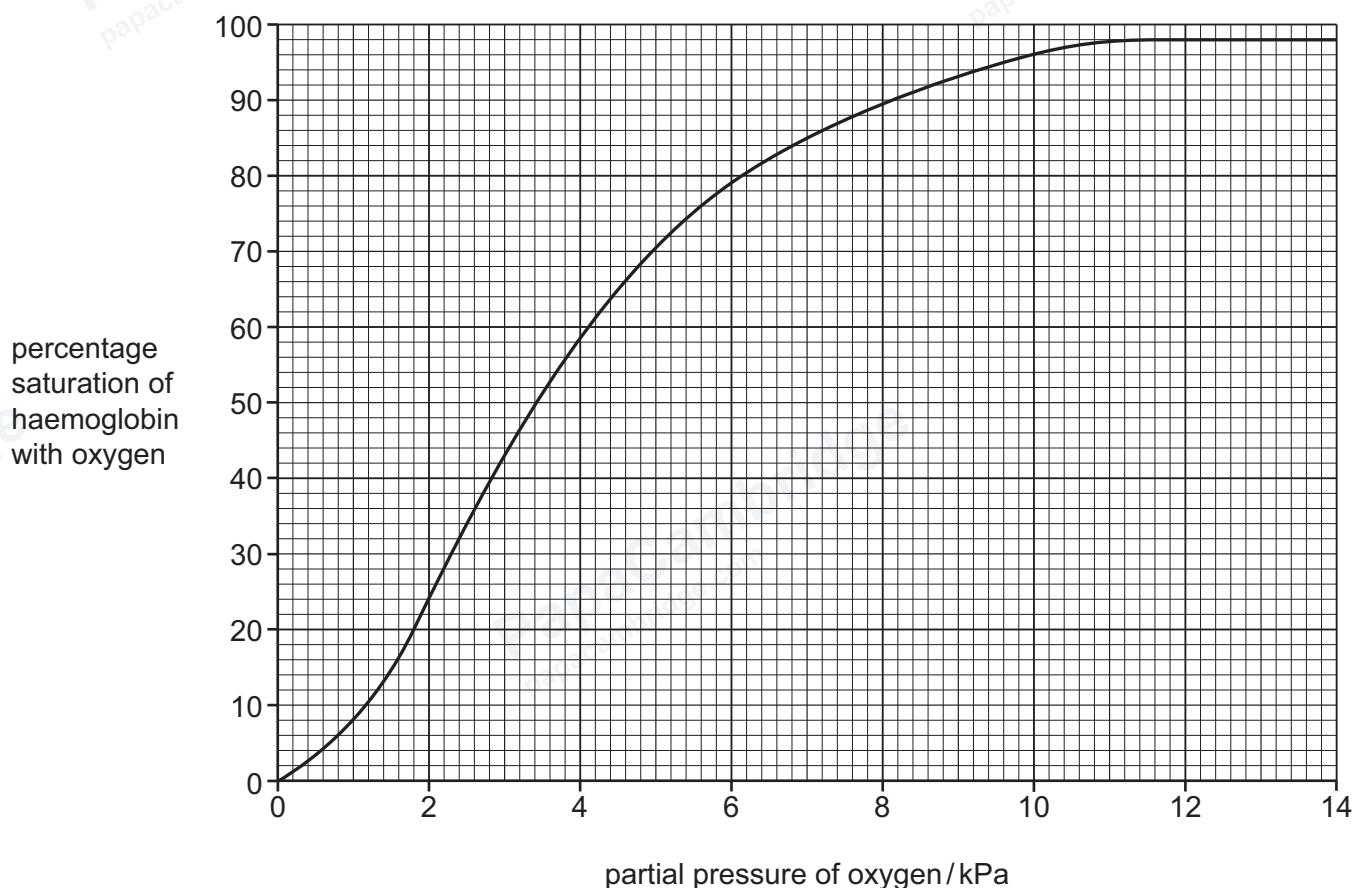


Figure 4.1

- (a) At sea level, the partial pressure of oxygen in the alveoli remains approximately the same at 13.2 kPa to 13.3 kPa.

With reference to Figure 4.1, explain why a partial pressure of 13.2 kPa to 13.3 kPa in the alveoli allows the uptake of oxygen into the alveolar capillaries.

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(b) The partial pressure of oxygen (pO_2) in a tissue can vary under different conditions.

- (i) With reference to Figure 4.1, calculate the difference in percentage saturation of haemoglobin with oxygen when a tissue pO_2 changes from 5.6 kPa to 2.4 kPa.

Show your working.

difference =% [2]

- (ii) Suggest **one** factor that could cause a change in pO_2 of a tissue.

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- (c) Blood leaving the capillary network of a body tissue is eventually returned to the right atrium of the heart.

Name the blood vessel that returns blood to the right atrium of the heart.

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[Total: 7]